ENTERPRISE SYSTEM AUDIT

Enterprise Upgrade Roadmap

Missing features, a competitive gap analysis, and a prioritized implementation plan.

Company Safari Zangu Ventures
Engagement Full-Codebase Enterprise Audit
Date 20 September 2026
Document 5 of 6
Classification Internal — Management & Engineering

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1. How to Read This Document

Everything in this report so far has been strictly code-verified. This document is different by necessity: the Competitor Gap Analysis section is a qualitative comparison based on general knowledge of how enterprise tour-operator platforms (TourPlan, Travel Studio, Wetu, and similar “Safari Portal”-style DMC systems) are typically built — not a line-by-line audit of their actual source code, which this engagement had no access to. Treat that section as directional, not verified fact, unlike every other finding in this document set.

2. Missing Features Summary

2.1 Operations Management — not built

Capability	Status	Evidence
Supplier / accommodation-contract management	✗ MISSING	grep -rniE 'supplier|contract' across the whole codebase returns exactly one hit, a marketing bio sentence on the About page — no table, no admin page, no data model
Guide assignment (as personnel, not a pricing multiplier)	✗ MISSING	No guides table exists. GUIDE_LANGUAGES config array is a pricing multiplier by spoken language, not a personnel/assignment system. All other 'guide' references are informal marketing copy
Vehicle assignment	■ PARTIAL	Vehicles are selected per-quotation (quotations.vehicle_id FK exists and is used), but there's no fleet-scheduling/conflict-detection layer — nothing prevents the same vehicle being assigned to two overlapping bookings
Driver assignment (distinct from vehicle cost)	✗ MISSING	driver_cost_day is purely a $ rate bundled into the vehicle's daily cost — no driver entity, no driver table, no assignment logic
Inventory / room-availability tracking	✗ MISSING	No availability/capacity tracking found anywhere; accommodations table has rates but no room-count or booking-conflict logic
Voucher generation	✗ MISSING	Zero matches for 'voucher' anywhere in the codebase
Operations calendar (distinct from reminder list)	✗ MISSING	admin/reminders.php is a table/list view over booking_reminders, not a calendar. Zero matches for 'calendar' anywhere
Departure management	■ PARTIAL	booking_reminders has a 'departure' reminder type (automated email), but no dedicated departure-day operations view (manifest, driver briefing, etc.)
Emergency / incident handling	✗ MISSING	Zero matches for 'emergency' or 'incident' anywhere in the codebase

2.2 Quotation Engine — built but disconnected or missing

Full detail in the Master Technical Specification; summarized here as roadmap input. Features marked “PARTIAL” generally have real schema/admin UI already built — they need calculation-layer wiring, not new data modeling.

Pricing capability	Status
Concession fees (distinct from park fees)	MISSING
Airport/airstrip fees or transfers	PARTIAL
Ground transfers as distinct line item	PARTIAL
Accommodation costing	PARTIAL
Vehicle costing (rate/fuel/driver/maint/insurance)	PARTIAL
Guide costs (distinct line item)	PARTIAL
Driver allowances (distinct from driver_cost_day)	MISSING
Fuel costs (own line item)	MISSING
Seasonal pricing (real multiplier)	MISSING (live) / PARTIAL (orphaned only)
Dynamic/demand-based pricing	MISSING
Profit margin / markup	MISSING
Multi-currency	MISSING
Child pricing	PARTIAL
Resident (Tanzanian) pricing	MISSING from calculation
EAC citizen pricing	MISSING
Single supplement	MISSING from any total

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3. Competitive Gap Analysis (Directional)

Reference systems named in the brief — TourPlan, Travel Studio, Wetu, and generic “Safari Portal”-style DMC platforms — are established, enterprise-grade tour-operator systems built over many years for multi-staff DMC/ground-handler operations. The comparison below reflects the feature categories those platforms are generally known to offer versus what exists in this codebase today; it is not a claim about their exact implementation.

Capability	Reference systems	Safari Zangu Ventures
Supplier/contract management with rate sheets	Standard feature in all four reference systems	Not built
Guide & driver rostering/assignment	Standard feature	Not built (driver cost is a flat vehicle-rate add-on only)

Capability	Reference systems	Safari Zangu Ventures
Room/seat inventory & allocation tracking	Standard feature	Not built
Voucher/exchange-order generation	Standard feature	Not built
Operations calendar (departures, resources)	Standard feature	Not built (a reminder list exists, not a calendar)
Resident/non-resident/EAC differential pricing	Standard for East Africa-focused systems	Schema exists, not applied in calculation
Seasonal rate calendars	Standard feature	Global multiplier exists but is not applied in the live pricing path
Multi-currency quoting	Standard feature	USD-only; GBP shown for display purposes only
Agent/DMC commission tracking	Standard feature	Built and working
Online booking + payment	Standard feature	Built and working (PayPal + bank transfer)
Customer self-service tracking	Less common even in reference systems	Built and working — a genuine differentiator for a company this size
Automated lifecycle emails (reminders, confirmations)	Standard feature	Built and working, with real-time send (upgraded this engagement)

4. Implementation Roadmap

Prioritized by business impact relative to effort — not strictly by technical severity. Several “Critical/High” items below are cheap fixes with outsized impact (e.g., wiring an already-built resident-pricing field into the calculator); several “Low” items are technically real but low-urgency at current data volume.

4.1 Critical Priority

Retire or reconnect the orphaned QuotationEngine.php CRITICAL

A second, more feature-complete pricing engine exists with zero real callers. Leaving it in place risks a future developer fixing a pricing bug in the wrong engine. Database changes: none. Backend changes: either delete QuotationEngine.php + api/quote.php, or migrate its season-multiplier and guide-cost logic into ItineraryQuoteBuilder and delete the duplicate. Frontend changes: none if retired; quote widget rewiring if merged. Business impact: prevents a real, silent pricing-bug risk.

```
EVIDENCE: api/quote.php has zero callers codebase-wide
```

4.2 High Priority

Apply resident/non-resident/EAC park-fee tiers to real calculations HIGH

Database changes: none — resident_adult/resident_child columns already exist and are already editable via admin/park_fees.php. Backend changes: add a nationality/residency selector to the booking and quotation flow, and update ItineraryQuoteBuilder's park-fee SELECT to branch on it. Frontend changes: one new form field. Business impact: correct pricing for Tanzanian and EAC-resident guests, a real revenue-accuracy fix, not just a feature addition.

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EVIDENCE: park_fees.resident_adult/resident_child exist, populated with $0.00
placeholders, never read by calculation code
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Apply seasonal multipliers to the live pricing path HIGH

Database changes: none — seasonal_rates table and admin CRUD already exist and work. Backend changes: wire ItineraryQuoteBuilder to look up and apply the multiplier by travel date, same logic already written (and unused) in QuotationEngine.php. Frontend changes: none required, though a season indicator on the quote would help sales. Business impact: currently every quote is priced identically regardless of season — a real margin/accuracy gap on high-season bookings.

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EVIDENCE: seasonal_rates table has real, correct data (peak=1.0/shoulder=0.85/low=0.75)
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Apply single-supplement and true child/accommodation-tier pricing HIGH

Database changes: none. Backend changes: read accommodations.single_supplement and child_rate in the accommodation costing step of ItineraryQuoteBuilder. Frontend changes: none required for MVP. Business impact: solo travelers and families are currently quoted incorrectly relative to the rates already on file.

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EVIDENCE: accommodations.single_supplement/child_rate exist, edited via admin, never read
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Fix the two confirmed broken admin features HIGH

Database changes: none. Backend/frontend changes: restore ADMIN_GUIDE.md or remove the broken link in admin/guide.php; either wire the AI-caption button to a real server-side Anthropic API call with a proper key, or remove it from poster_maker.php. Business impact: low usage-frequency but currently a live, user-visible bug.

`EVIDENCE: admin/guide.php:31 confirmed live warning; admin/poster_maker.php's client-side Anthropic call has no auth header`

4.3 Medium Priority

Add markup/margin as a real pricing step MEDIUM

Database changes: none — package_tier/operational_cost/selling_price/profit_margin_percent columns were added this engagement but are not yet wired to calculation. Backend changes: add an explicit markup percentage step between raw itinerary cost and price_1pax/quotation totals, and populate operational_cost from the existing cost_breakdown data. Frontend changes: a markup-% field in package_edit.php. Business impact: currently price_1pax is exactly cost+VAT with no margin step — profitability is invisible without this.

`EVIDENCE: vw_package_profitability view (added this engagement) currently shows 100% margin for every package because operational_cost is unset`

Standardize database collation and consolidate duplicated SMTP code MEDIUM

Database changes: one migration to unify all tables on utf8mb4_unicode_ci, removing the ad-hoc COLLATE workaround in reminders.php. Backend changes: refactor the 5 remaining duplicated PHPMailer setup blocks to call the already-built Mailer::sendRow(). Frontend changes: none. Business impact: maintainability, not user-facing.

`EVIDENCE: sql/baseline_schema.sql collation mismatch; SMTP block duplicated in 6 locations`

Add pagination to the 13 remaining unbounded admin lists MEDIUM

Database changes: none. Backend changes: apply the existing bookings.php pagination pattern to accommodations, activities, itineraries, packages, park_fees, seasonal_rates, team_members, users, vehicles, email_templates, commissions, reminders, contact_messages. Frontend changes: page-number UI, already built once and reusable. Business impact: prevents future page-load degradation as the catalog grows.

`EVIDENCE: 13 admin list pages confirmed with no LIMIT or true pagination`

4.4 Low Priority

Remove confirmed dead code LOW

Database changes: none. Backend changes: delete api/mailer.php, BookingConfirmationMail.php, CustomSafariAckMail.php, and 8 unused model methods, all confirmed to have zero call sites. Frontend changes: none. Business impact: codebase hygiene only.

`EVIDENCE: See Master Technical Specification, Code Quality section, for the full list`

Add explicit image dimensions sitewide LOW

Frontend-only change: add width/height attributes to the 26 image tags currently missing them, preventing layout shift. Business impact: minor UX polish.

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